# Assignment-1 : Real Estate Market Insights

# Task 1 — Load & Inspect the Dataset

1. Load the dataset into your notebook.
2. Display the first 5 and last 5 rows of the dataset.
3. Print dataset shape (rows, columns).
4. Print dataset information using .info().
5. Identify basic data types of all columns.

## **Task 2 — Clean the Dataset**

1. Identify and count missing values in each column.
2. Drop column(s) that are irrelevant for price analysis (example: society or unnamed columns).
3. Handle missing values in numeric columns such as bath, balcony, etc. (use either dropna() or fillna() depending on your reasoning).
4. Convert total_sqft to numeric — handle values like "2100 - 2850" by converting them to an average or a single number.
5. Remove duplicate rows.

6. Reset the DataFrame index after cleaning

# TASK 3

# Task - 3 Solution:

```
# 1) How many unique locations?

print("1) Number of unique locations:", df['location'].nunique())
```

```python
# 2) Average house price per location

avg_price = df.groupby('location')['price'].mean().sort_values(ascending=False)

print("\n2) Average price per location (Top 10):")

print(avg_price.head(10))


# 3) Location with highest average price

top_loc = avg_price.idxmax()

top_price = avg_price.max()

print(f"\n3) Highest average price location: {top_loc} — {top_price:.2f} lakhs")


# 4) Correlation between sqft, bath, price

corr = df[['total_sqft', 'bath', 'price']].corr()

print("\n4) Correlation:\n", corr)


# 5) Interpretation (simple logic)

corr_sqft_price = corr.loc['total_sqft', 'price']

print("\n5) Interpretation:")


if corr_sqft_price > 0.7:

    print("• Strong positive correlation → Larger houses generally have higher prices.")

elif corr_sqft_price > 0.4:

    print("• Moderate correlation → Larger houses often cost more, but not always.")

else:

    print("• Weak correlation → Size alone does NOT determine price.")
```

# Task 4 — Data Visualization

Use the **specific visualization types mentioned** for each question.

1. **Price Distribution:**
   - Plot the distribution of the price column using a **Histogram + KDE curve**.
2. **Relationship Between Area and Price:**
   - Visualize the relationship between total_sqft and price using a **Scatter Plot**.
3. **Effect of Bathrooms on Price:**
   - Show how bath count affects house prices using a **Box Plot**.
4. **Top 10 Most Expensive Locations:**
   - Plot the **Top 10 locations** with highest average price using a **Bar Chart**.
5. **Correlation Between Numeric Columns:**
   - Create a **Heatmap** to visualize correlations among numeric features (price, total_sqft, bath, balcony, etc.).